

Computational Investigation of an Active Dihedral Vector Control Mechanism for Decoupled Yaw Authority in Motorsport Applications

A 3D Flow Visualization and CFD Study of a Proposed Aero-Vectoring Concept

Danial Moghadam – *M.Sc. in Aerodynamics*

Independent Research

dan998m@gmail.com

Portfolio Repository: github.com/DanMoghadam-Aero

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Abstract

This study evaluates the Active Dihedral Vector Control (ADVC), a proposed aerodynamic yaw-control architecture for motorsport applications. Unlike conventional systems that generate yaw primarily through asymmetric drag, the ADVC uses independently actuated, full-chord dihedral outboard wing sections to redirect part of the local aerodynamic loading laterally and produce a yaw moment about a proposed vehicle centre of gravity. A three-dimensional steady Reynolds-averaged Navier–Stokes (RANS) computational fluid dynamics sweep comprising 14 cases examined neutral and single-sided deployments of up to 15° under imposed sideslip angles of up to 5° . Integrated aerodynamic loads and qualitative flow structures were evaluated across the resulting control envelope.

At zero sideslip, deployment produced a monotonic lateral-force increment while retaining approximately 98.4% of the isolated wing assembly’s baseline downforce. The corresponding yaw-moment increments were consistent with the measured side force acting through a plausible longitudinal lever arm of 1.5–1.8 m, supporting, but not uniquely proving, side-force-dominated yaw generation. The resolved longitudinal-force coefficient increment remained close to zero, indicating no discernible drag penalty at the present numerical fidelity; however, sub-percent drag changes remain within the uncertainty of the steady-RANS analysis. The mechanism also retained bidirectional and approximately linear yaw authority under the investigated sideslip conditions, while the accompanying roll- and pitch-moment increments remained smaller than the primary yaw response. Within the limitations of the isolated-wing geometry and preliminary numerical methodology, these findings support the aerodynamic plausibility of the ADVC concept and justify subsequent experimental, mesh-refined, and full-vehicle investigations.

Keywords: Active Aerodynamics; Yaw Vectoring; Computational Fluid Dynamics (CFD); Dihedral Actuation; Cross-Axis Decoupling; Motorsport Aerodynamics; NACA 4415.

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1 Motivation, Research Question, and Scope

1.1 Engineering Motivation and Problem Definition

Aerodynamic control surfaces in high-performance vehicles are commonly employed to regulate downforce and longitudinal drag. Their application to direct aerodynamic yaw control, however, presents a different vehicle-dynamics opportunity. Conventional aerodynamic yaw-control methods frequently employ asymmetric drag, which generates a yaw moment but inherently introduces a parasitic drag penalty.

The present concept investigates an alternative approach: using active aerodynamic surfaces to generate a directed lateral force behind a hypothetical vehicle centre of gravity (CG). Acting through this longitudinal lever arm, the lateral force produces a yaw-moment contribution that could subsequently be evaluated as a control input in coupled vehicle-dynamics studies of corner-entry or stability-control applications. This motivates the investigation of a side-force-dominated yaw mechanism capable of limiting undesirable changes to the remaining aerodynamic loads.

1.2 Theoretical Context and Concept Definition

High-performance vehicles have explored several approaches to active aerodynamic load management. Some concepts use whole-wing tilting mechanisms to redirect the aerodynamic force laterally, while others employ asymmetric flaps as airbrakes to generate yaw through differential drag. Such systems have previously been investigated for vehicle handling, load redistribution, and yaw-stability applications [4, 5, 6].

The Active Dihedral Vector Control (ADVC) mechanism investigated in this study provides a conceptual testbed for an alternative geometric approach. It consists of two independently articulated, full-chord outer sections adjoining a fixed central rear-wing section. Dihedral deployment of either outer section redirects part of its local aerodynamic loading in the spanwise direction. The resulting lateral force acts through a longitudinal lever arm relative to the proposed vehicle CG, thereby generating a yaw moment without relying primarily on asymmetric drag or whole-wing rotation.



Figure 1: Examples of production active-aerodynamic systems used as design context. Left: Zenvo TSR-S Centripetal Wing (image source: Top Gear [9]). Right: Pagani Huayra (image source: CarBuzz [10]). These examples provide contextual comparison only and are not aerodynamically equivalent to the proposed ADVC concept.

1.3 Research Question and Evidence Boundary

The principal research question is therefore:

Can independently actuated dihedral outer sections generate measurable, bidirectional aerodynamic yaw authority primarily through lateral side force, while limiting changes in longitudinal force and

downforce and maintaining smaller accompanying roll- and pitch-moment increments than the primary yaw-moment response?

The analysis is restricted to an isolated wing assembly evaluated under prescribed freestream and sideslip conditions. It quantifies the incremental aerodynamic forces and moments produced by the active surfaces but does not model a complete vehicle chassis. Consequently, coupled vehicle handling, transient actuation response, structural feasibility, and wing-body interaction remain outside the scope of the present study. The conclusions are therefore limited to the aerodynamic feasibility and comparative load trends of the isolated ADVC mechanism.

2 Geometry, Kinematics, and Coordinate Definitions

2.1 Wing Architecture

The investigated device is based on an inverted NACA 4415 wing profile with a nominal chord of

$$c = 0.30 \text{ m.} \quad (1)$$

The original wing was extruded to a total span of 1.50 m and subsequently divided into a fixed central section and two independently actuated outer sections. The central section has a span of 0.60 m, while each outer section has a span of 0.45 m.

Each outer section was translated outward by 0.020 m to provide clearance during deployment and to simplify the meshing process. The resulting nominal undeployed tip-to-tip span is therefore

$$b_{\text{ref}} = 0.45 + 0.020 + 0.60 + 0.020 + 0.45 = 1.54 \text{ m,} \quad (2)$$

excluding any additional contribution from the thickness of the endplates which were negligible. The moving parts are full-chord outer wing sections rather than conventional trailing-edge flaps. Each endplate is attached to its corresponding outer section and therefore moves with it.

Two swan-neck supports connect the fixed central wing section to the conceptual vehicle structure. Their spanwise positions are approximately ± 0.20 m from the wing centreline. These dimensions were selected as generic values suitable for the present proof-of-concept investigation and were not obtained through structural or aerodynamic optimization.

The principal dimensions and reference quantities are summarized in Table 1.

Table 1: Principal wing dimensions and aerodynamic reference quantities.

Quantity	Value or convention
Airfoil profile	Inverted NACA 4415
Nominal chord, c	0.30 m
Fixed central-section span	0.60 m
Outer-section span	0.45 m per side
Clearance gap	0.020 m per side
Original extrusion span	1.50 m
Reference span, b_{ref}	1.54 m, including gaps
Reference area, S_{ref}	0.45 m^2
Swan-neck lateral positions	Approximately ± 0.20 m
Endplate dimensions	$0.35 \text{ m} \times 0.15 \text{ m}$

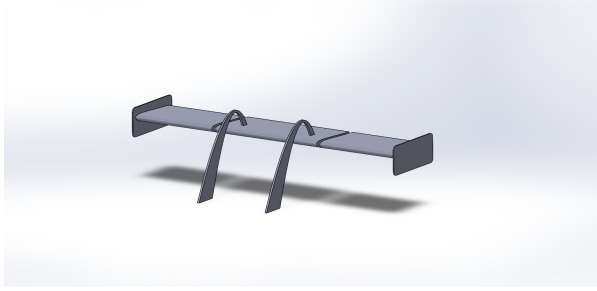
The reference lifting area is defined as

$$S_{\text{ref}} = c(b_{\text{centre}} + 2b_{\text{outer}}) = 0.30(0.60 + 2 \times 0.45) = 0.45 \text{ m}^2. \quad (3)$$

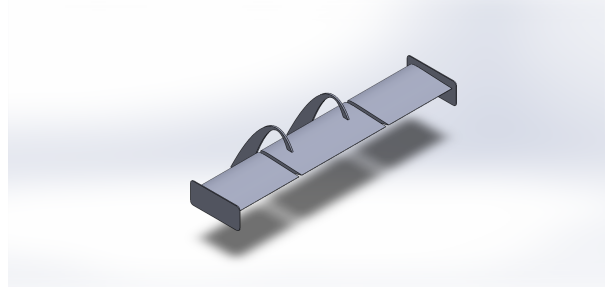
This area excludes the clearance gaps, endplates, and swan-neck mounts. It is held constant for every configuration and does not represent the instantaneous horizontal projected area of the deployed wing. The reference span is likewise held constant when nondimensionalizing the roll and yaw moments.

2.2 CAD Geometry and Principal Views

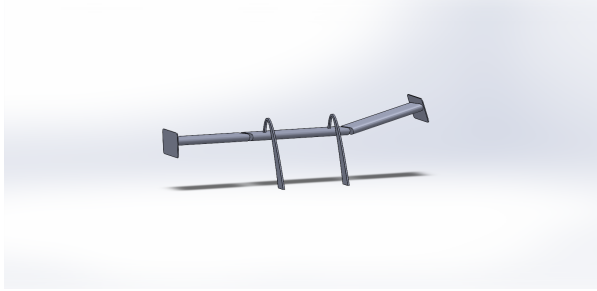
The principal CAD views are presented below showcasing baseline geometry and several deployed scenarios.



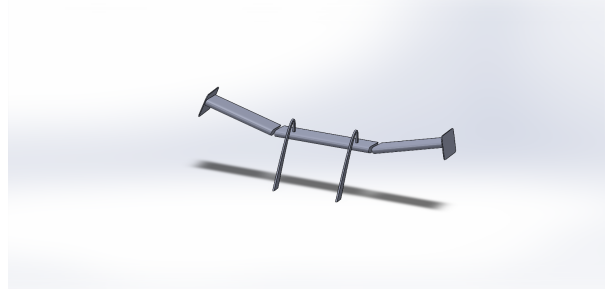
(a) Neutral baseline



(b) Neutral baseline (Isometric view)



(c) 15° Left flap deployment



(d) Both dihedral flaps deployed to 15°

Figure 2: Kinematic architecture of the ADVC mechanism. (a) and (b) establish the 0.60 m fixed central section and 0.45 m outer flaps in the neutral state. (c) showcases independent deployment and (d) illustrates both dihedral sections in a deployed state (for illustration purpose only, otherwise simultaneous deployment was not investigated).

2.3 Actuation and Kinematic Variables

The left and right outer-section deployment angles are denoted by δ_L and δ_R , respectively. Both SolidWorks parameters were manipulated in such a way that positive value always represents upwards deployment.

The nominal neutral CAD inputs exported are

$$\delta_L = \delta_R = 0.01^\circ. \quad (4)$$

The small nonzero value was used as the nominal neutral position in the parameterized workflow. Only one outer section was actuated in each deployed configuration. When the left section was deployed, the right section remained at its nominal neutral value of 0.01° . Similarly, when the

right section was deployed, the left section remained at 0.01° . The present design matrix therefore contains:

- a neutral configuration;
- upward deployment of the left outer section; and
- upward deployment of the right outer section.

Simultaneous deployment and downward deployment were not investigated.

2.4 Coordinate System and Aerodynamic Load Convention

The coordinate system and aerodynamic-load definitions used throughout this study are illustrated in Figure 3. The axes are defined as follows:

- $+X$ points longitudinally from the front toward the rear of the vehicle and follows the downstream airflow direction;
- $+Y$ points vertically upward; and
- $+Z$ points spanwise toward the left (The driver's left side that is).

The integrated force components are F_x , F_y , and F_z . Positive downforce is defined by

$$F_{\text{down}} = -F_y. \quad (5)$$

The corresponding moment components are

$$M_x = \text{roll moment}, \quad M_y = \text{yaw moment}, \quad M_z = \text{pitch moment}. \quad (6)$$

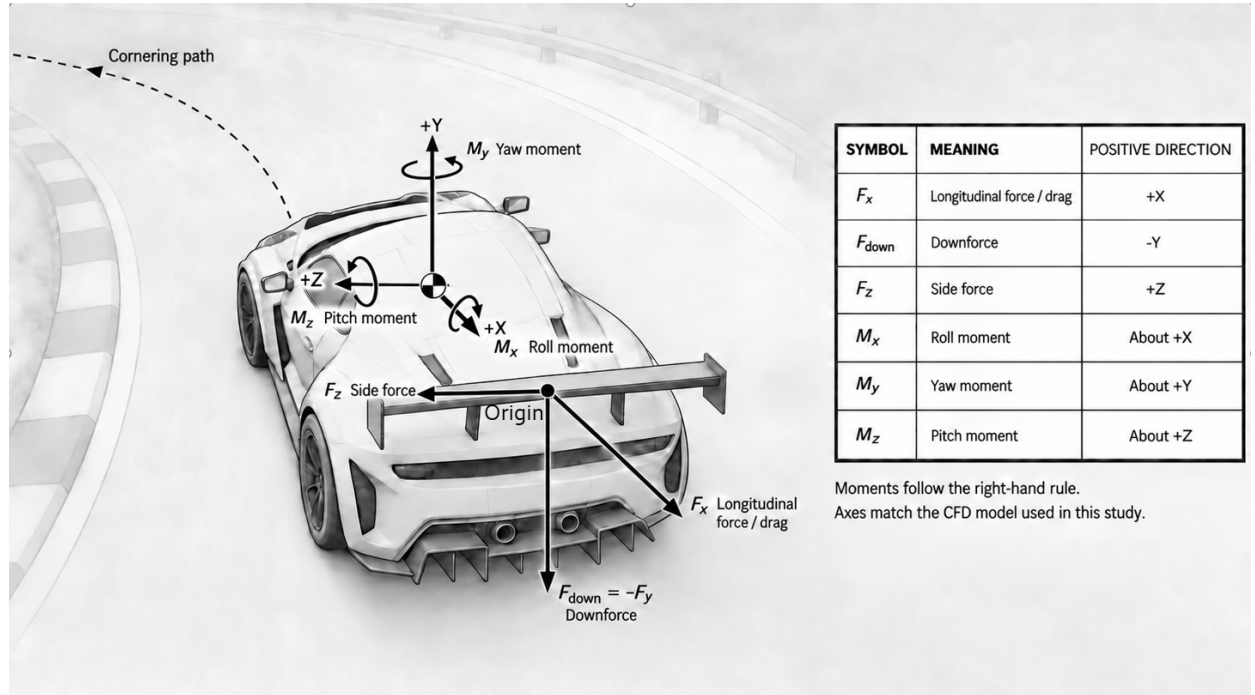


Figure 3: Conceptual illustration of the coordinate system and aerodynamic load conventions used throughout the study (with the origin shown at the centerline of the wing Leading Edge). The moments are defined about the proposed vehicle centre of gravity and follow the right-hand rule. The illustrated vehicle provides physical context and was not included in the CFD domain.

The force vectors in Figure 3 are schematic. The forces reported by Fluent are integrated over the selected wall surfaces of the complete simulated wing assembly. Their arrows should therefore be interpreted as definitions of direction rather than measured points of application.

2.5 Moment Reference Centre

The yaw and pitching moments exported from Ansys Fluent were evaluated about the proposed vehicle centre of gravity. The specified reference coordinates are

$$x_{CG} = -1.5 \text{ m}, \quad y_{CG} = -0.6 \text{ m}, \quad z_{CG} = 0. \quad (7)$$

For an elemental surface-force contribution $d\mathbf{F}$ acting at position $\mathbf{x}$, define

$$\mathbf{r} = \mathbf{x} - \mathbf{x}_{CG}. \quad (8)$$

The total moment about the proposed centre of gravity is

$$\mathbf{M}_{CG} = \int_A \mathbf{r} \times d\mathbf{F}. \quad (9)$$

Its components are

$$M_x = \int_A (r_y dF_z - r_z dF_y), \quad (10)$$

$$M_y = \int_A (r_z dF_x - r_x dF_z), \quad (11)$$

$$M_z = \int_A (r_x dF_y - r_y dF_x). \quad (12)$$

Both pressure and viscous surface friction contribute to the integrated forces and moments.

Equation (11) shows that the yaw moment contains contributions from both the longitudinal force i.e., drag and the side force, induced by dihedral deployment. So it is necessary to decompose these two forces later and evaluate the contribution of each in order to verify the efficacy of the ADVc mechanism.

3 Computational Methodology and Freestream Parameters

3.1 Freestream Conditions and Slip-Angle Induction

The computational domain was initialized to represent a simplified freestream condition. The freestream fluid was modeled as incompressible air with a constant density of $\rho = 1.225 \text{ kg/m}^3$ and a dynamic viscosity of $\mu = 1.7894 \times 10^{-5} \text{ kg/ms}$. A uniform freestream velocity magnitude of $U_\infty = 40 \text{ m/s}$ was prescribed at the domain inlet, corresponding to a dynamic pressure of $q_\infty = 980 \text{ Pa}$ and a chord-based Reynolds number of $Re_c \approx 8.22 \times 10^5$.

To isolate the pure yaw and roll derivatives generated exclusively by the ADVc mechanism, the baseline wing was evaluated at a nominal 0° angle of attack. This parameter was fixed so as not to add another variable.

To evaluate the robustness of the active surfaces during active cornering, lateral crossflow was introduced to simulate vehicle slip angle (β). Rather than physically rotating the CAD geometry within the domain—which alters the mesh topology and introduces grid-variance errors—slip angles

were prescribed by decomposing the inlet velocity vector into longitudinal (U_x) and lateral (U_z) components. Alongside the nominal 0° baseline, crossflow cases were evaluated at $\beta \approx 2.5^\circ$ and $\beta \approx 5.0^\circ$.

3.2 Domain and Boundary Conditions

To eliminate artificial flow acceleration and solid blockage effects, the computational domain was set up as a rectangular Cartesian enclosure around the Active Dihedral Vector Control (ADVC) geometry. The fluid domain was extended asymmetrically to adequately capture the upstream onset flow and the turbulent downstream wake structures.

The rectangular enclosure was generated in Ansys DesignModeler using non-uniform cushion distances applied outward from the bounding box of the complete wing assembly. The reported dimensions therefore represent clearances from the farthest point on the geometry rather than coordinates measured from the global origin. The inlet was positioned 2.0 m, or approximately $6.7c$, upstream ($-X$) of the most upstream point of the geometry, while the outlet was positioned 4.0 m, or approximately $13.3c$, downstream ($+X$) of the most downstream point. The upper and lower boundaries ($\pm Y$) were each offset by 1.5 m ($5c$), and the lateral boundaries ($\pm Z$) extended 1.0 m ($3.3c$) beyond the corresponding outermost spanwise locations.

The boundary conditions were mapped as follows:

- **Inlet:** A velocity inlet was applied at the upstream boundary ($-X$).
- **Outlet:** A zero-gauge static pressure outlet was applied at the downstream boundary ($+X$).
- **Outer Enclosure Boundaries:** The top, bottom, and lateral boundaries were defined as zero-shear slip walls to simulate an unbounded freestream environment without inducing artificial boundary layer growth along the domain limits.
- **Aerodynamic Surfaces:** The ADVC geometry, explicitly divided into 3 parts, (mainly the fixed center section, left and right flaps) was designated as a no-slip wall.

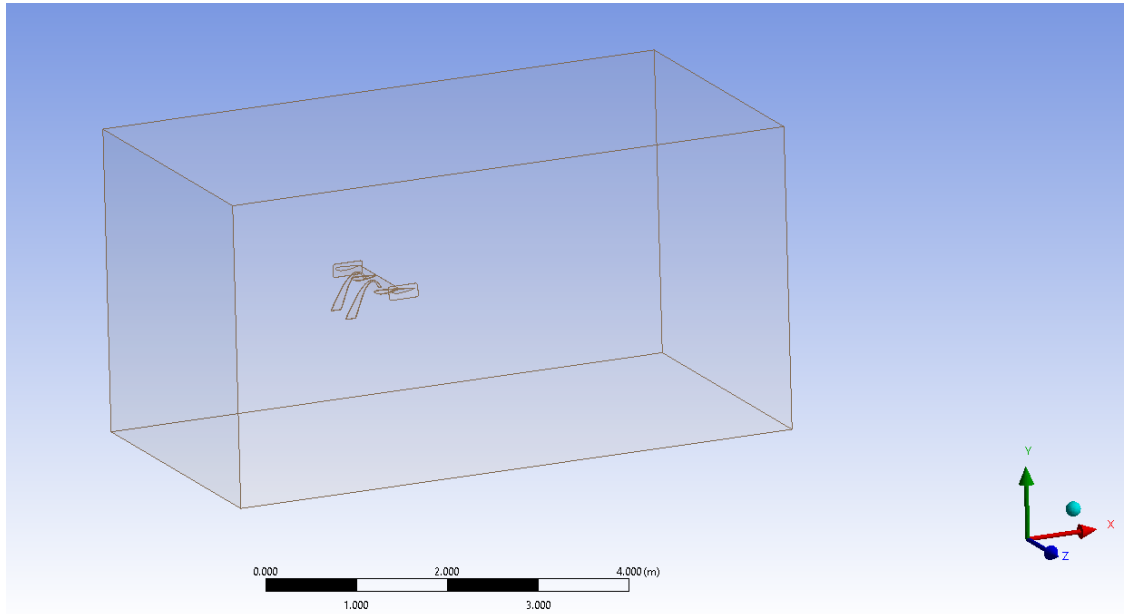


Figure 4: Computational fluid domain surrounding the ADVC geometry. The rectangular Cartesian enclosure features extended asymmetrical downstream bounds to capture turbulent wake structures, with symmetry (slip-wall) conditions applied to the outer boundaries to prevent artificial solid blockage.

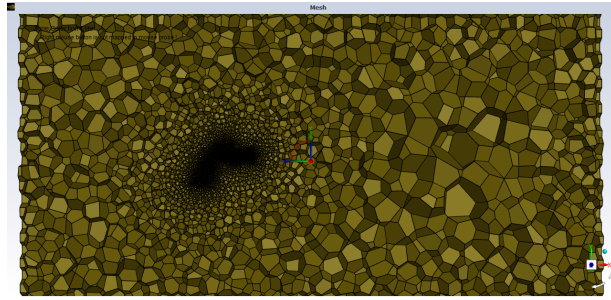
3.3 Mesh and near-wall resolution

Discretization was done using Ansys Fluent Meshing, utilizing an unstructured polyhedral volume mesh. Polyhedral elements were selected due to their superior geometric adaptability and numerical efficiency, offering faster convergence and reduced numerical diffusion compared to standard tetrahedral meshes.

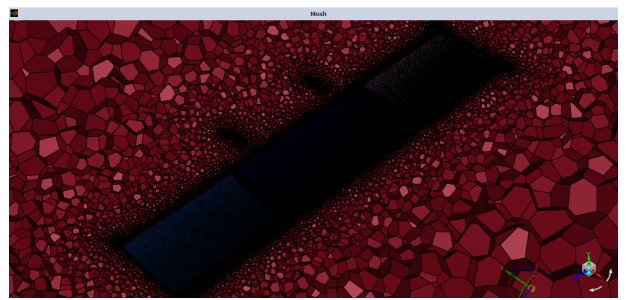
The surface mesh was driven by a Curvature and Proximity size function to accurately capture the high-gradient regions around the leading edges and flap articulation gaps. The global surface mesh size was bounded between a minimum of 0.5 mm and a maximum of 0.5 m, utilizing a normal angle limit of 18° , a minimum of 4 cells per gap, and a surface growth rate of 1.2. To ensure acceptable resolution over the lifting surfaces, a localized face sizing of 4.0 mm was enforced across all aerodynamic zones.

Near-wall velocity gradients and boundary layer physics were treated using prismatic inflation layers extruded normal to the aerodynamic surfaces. A smooth-transition offset method was used to generate 3 consecutive prism layers with a transition ratio of 0.272 and a uniform growth rate of 1.2, transitioning smoothly into the core volume grid.

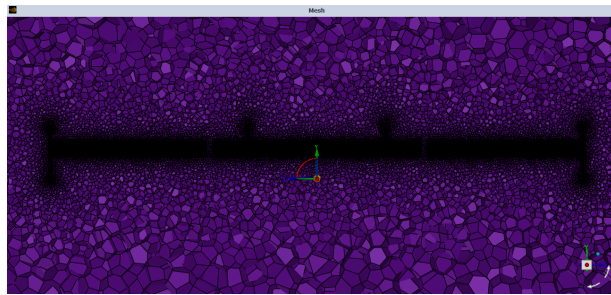
The core volume was populated with polyhedra up to a maximum cell length of 0.5 m. The final computational grid comprised 1,633,519 polyhedral cells, 9,406,820 faces, and 7,310,147 nodes. This high node-to-cell ratio is characteristic of polyhedral type meshes. All simulations were performed on my personal laptop (11th Gen Intel Core i7-11370H, 16 GB RAM) without access to institutional HPC resources; this constraint directly informed the mesh sizing, near-wall layer count, and iteration limit described in the following sections.



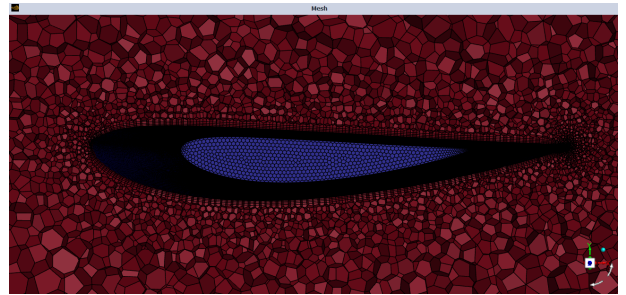
(a) Global polyhedral volume on $Z = 0$ - Side view.



(b) Surface mesh detail



(c) Front view



(d) Near-wall detail showing the 3 inflation layers.

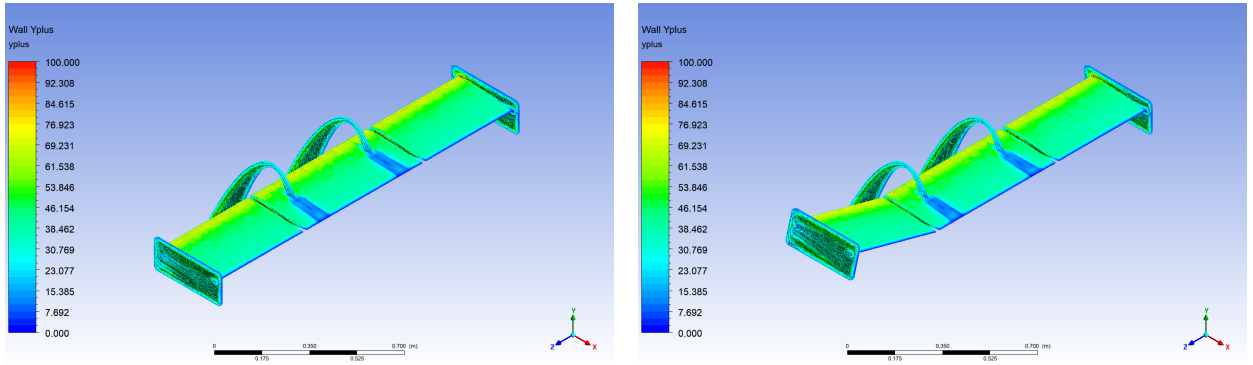
Figure 5: Computational grid mesh for the ADVC mechanism. The unstructured polyhedral mesh utilizes localized face sizing (4.0 mm) and smooth-transition boundary layers to resolve the aerodynamic gradients while sticking to total volume cell and computational constraints.

3.4 Solver Settings, Turbulence Modeling and Near-Wall Treatment

The closure problem of RANS was mitigated by utilizing the $k - \omega$ Shear Stress Transport (SST) turbulence model. The SST formulation was deliberately chosen for its hybrid nature, leveraging the

robustness of the $k - \epsilon$ model in the freestream while transitioning to the $k - \omega$ formulation in the near-wall region for superior prediction of adverse pressure gradients and flow [7]. To prevent the anomalous over-prediction of turbulent kinetic energy typical of two-equation models at stagnation points, the turbulent production limiter was enabled.

To remain within the available memory (see section 3.3), the near-wall mesh was limited to three prismatic layers, producing a mixed near-wall resolution with a maximum y^+ of approximately 78. Fluent’s SST implementation blends near-wall formulations across different y^+ regimes; however, the use of only three prism layers limits resolution of the boundary-layer velocity profile and should be regarded as a computational compromise rather than a grid-independent near-wall treatment. The y^+ contours in Figure 6 are therefore presented to document the achieved wall resolution and its similarity between configurations, not as direct evidence of boundary-layer attachment.



(a) Baseline configuration (0° deployment).

(b) Extreme yaw configuration (15° left deployment).

Figure 6: Wall- y^+ distributions for the baseline and 15° left-flap configurations at zero sideslip. Both cases exhibit similar near-wall-resolution patterns, with a maximum y^+ of approximately 78. The contours document mesh resolution only and are not used independently to infer flow attachment.

3.5 Solver Configuration and Convergence Behavior

The numerical solution was driven by a steady-state, pressure-based solver utilizing a Coupled pressure-velocity algorithm and second-order schemes for pressure, momentum and turbulence.

Due to computational constraints, solver execution was capped at a strict limit of 300 iterations per design point. During iteration, the continuity and momentum residuals reached bounded plateaus while integrated force monitors became approximately stationary. The solutions were therefore treated as monitor-converged for this preliminary comparative study, although the residual behavior indicates remaining numerical uncertainty.

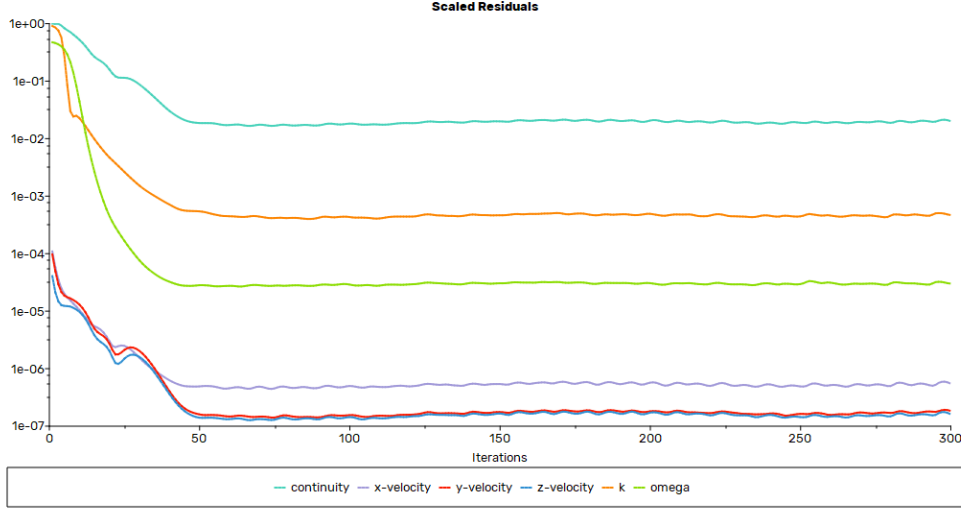
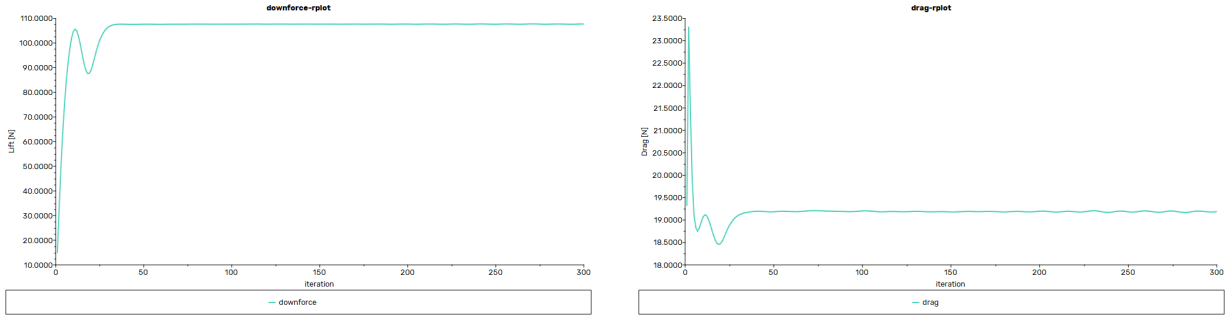


Figure 7: Continuity, momentum, and turbulence residual history for the extreme 15° left-deployment case (DP9).



(a) Asymptotic flattening of total downforce.

(b) Asymptotic flattening of longitudinal drag.

Figure 8: Solver force monitor behavior over the 300-iteration limit. Both the independent downforce (a) and drag (b) monitors confirm practical macroscopic convergence and bounded stability, supporting preliminary comparative load extraction.

4 Data Reduction and Mathematical Definitions

4.1 Incremental Loads and Matching Baselines

In order to isolate the aerodynamic influence of the Active Dihedral Vector Control (ADVC) deployment, all active results are evaluated as matched-baseline increments. Because only single-side actuation was investigated in this study, the actuated deployment angle is denoted simply as δ . For any exported scalar quantity Q , the active control increment is defined as:

$$\Delta Q(\delta, \beta) = Q(\delta, \beta) - Q(\delta = 0, \beta) \quad (13)$$

This subtraction removes the corresponding baseline crossflow load from the reported control increment, providing an estimate of the load change associated with flap deployment.

4.2 Freestream Quantities

For cases involving an induced slip angle (β), the local freestream velocity and dynamic pressure were derived from the prescribed inlet velocity components (U_x, U_z):

$$U_\infty = \sqrt{U_x^2 + U_z^2}, \quad \beta = \arctan\left(\frac{U_z}{U_x}\right), \quad q_\infty = \frac{1}{2}\rho U_\infty^2 \quad (14)$$

4.3 Nondimensionalization and Retention Metrics

All force and moment coefficients were non-dimensionalized against the fixed geometric parameters of the baseline wing to maintain a consistent reference standard across all deployment angles. While classical aerodynamic convention typically normalizes pitch using the reference chord (c), this study adopts a uniform non-dimensionalization strategy. To facilitate a rigorous 1:1 physical comparison of cross-axis torques acting upon the vehicle center of gravity, all rotational moments (roll, yaw, and pitch) utilize the identical reference span ($b_{\text{ref}} = 1.54$ m) as their characteristic length.

$$\text{Longitudinal Force: } C_x = \frac{F_x}{q_\infty S_{\text{ref}}} \quad (15)$$

$$\text{Downforce: } C_{\text{down}} = \frac{F_{\text{down}}}{q_\infty S_{\text{ref}}} \quad (16)$$

$$\text{Side Force: } C_{\text{side}} = \frac{F_z}{q_\infty S_{\text{ref}}} \quad (17)$$

$$\text{Roll Moment: } C_{M_x} = \frac{M_x}{q_\infty S_{\text{ref}} b_{\text{ref}}} \quad (18)$$

$$\text{Yaw Moment: } C_{M_y} = \frac{M_y}{q_\infty S_{\text{ref}} b_{\text{ref}}} \quad (19)$$

$$\text{Pitch Moment: } C_{M_z} = \frac{M_z}{q_\infty S_{\text{ref}} b_{\text{ref}}} \quad (20)$$

To evaluate the aerodynamic trade-offs of the ADVC mechanism, downforce retention (R_{down}) is expressed as a percentage ratio relative to the matching baseline configuration:

$$R_{\text{down}} = 100 \left(\frac{F_{\text{down}}}{F_{\text{down},0}} \right) \quad (21)$$

5 Results: Aerodynamic Control Authority and Cross-Axis Coupling

5.1 Force Decoupling and Actuation Linearity

The primary objective of the Active Dihedral Vector Control (ADVC) is to generate steering yaw via lateral side-force, explicitly avoiding the highly inefficient, asymmetric parasitic drag utilized by contemporary hypercar airbrakes. To validate this mechanical isolation in clean air ($\beta = 0^\circ$), the total yaw moment coefficient magnitude ($|\Delta C_{M_y}|$) was plotted alongside the estimated side-force-induced yaw moment ($|\Delta C_{M_y, F_z}|$).

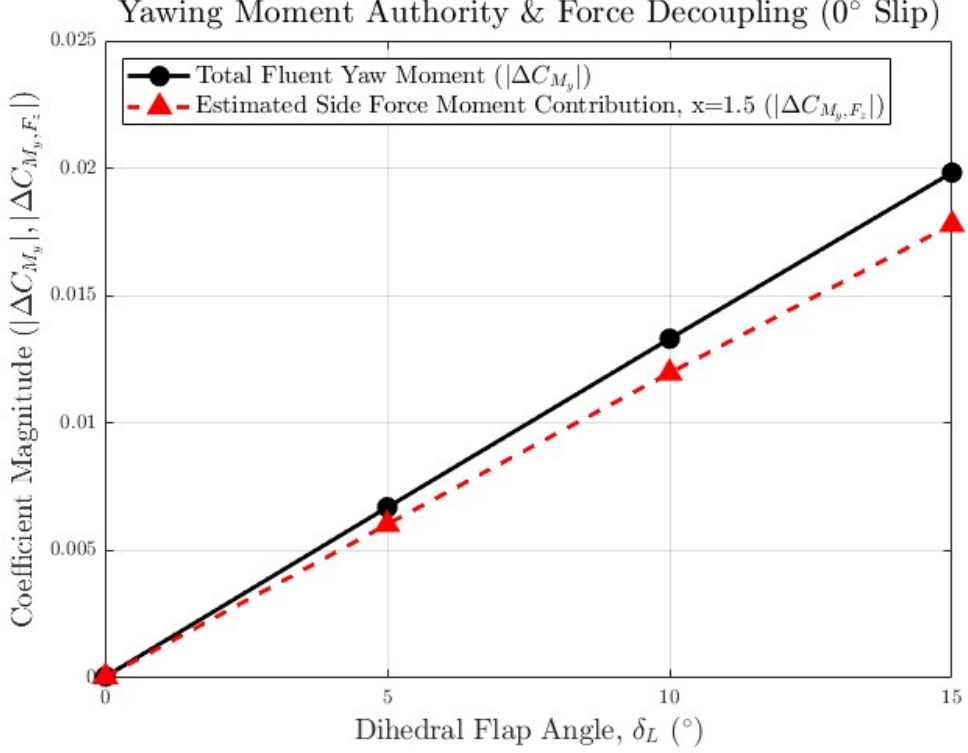


Figure 9: Yawing moment authority and force decoupling at 0° slip. The tight correlation between the total generated yaw moment ($|\Delta C_{M_y}|$) and the estimated side-force moment contribution ($|\Delta C_{M_y, F_z}|$) supports the idea that the majority of the yaw moment is generated by the lateral side force.

As illustrated in Figure 9, the mechanism exhibits approximate linearity within the sampled steady dataset. Moreover, the control authority scales proportionally to the deployment angle. This approximately linear response may simplify interpolation within the sampled steady-state range, although transient response and controller design were not investigated.

Furthermore, the two curves maintain a tight correlation; at the maximum 15° deployment, the side-force moment contribution ($|\Delta C_{M_y, F_z}| = 0.0178$) accounts for most of the total yaw moment magnitude ($|\Delta C_{M_y}| = 0.0199$).

The robustness of the observed trend is supported by its monotonicity and internal force–moment consistency. Across the 0°–15° deployment range, the incremental side force and yaw moment vary approximately linearly with deployment angle, with $R^2 > 0.9998$ for both quantities. Moreover, the diagnostic ratio $-\Delta M_y / \Delta F_z$ remains between 1.666 and 1.671 m, consistent with the geometric longitudinal lever-arm range.

5.2 Bidirectional Control Envelope and Crossflow Robustness

As a preliminary aerodynamic requirement, the mechanism should retain yaw authority under sideslip. Figure 10 illustrates the maximum bidirectional control envelope across crossflow angles up to $\beta = 5^\circ$. At zero sideslip, the undeployed geometry, computational domain, and boundary conditions are mirror-symmetric about the wing centreplane.

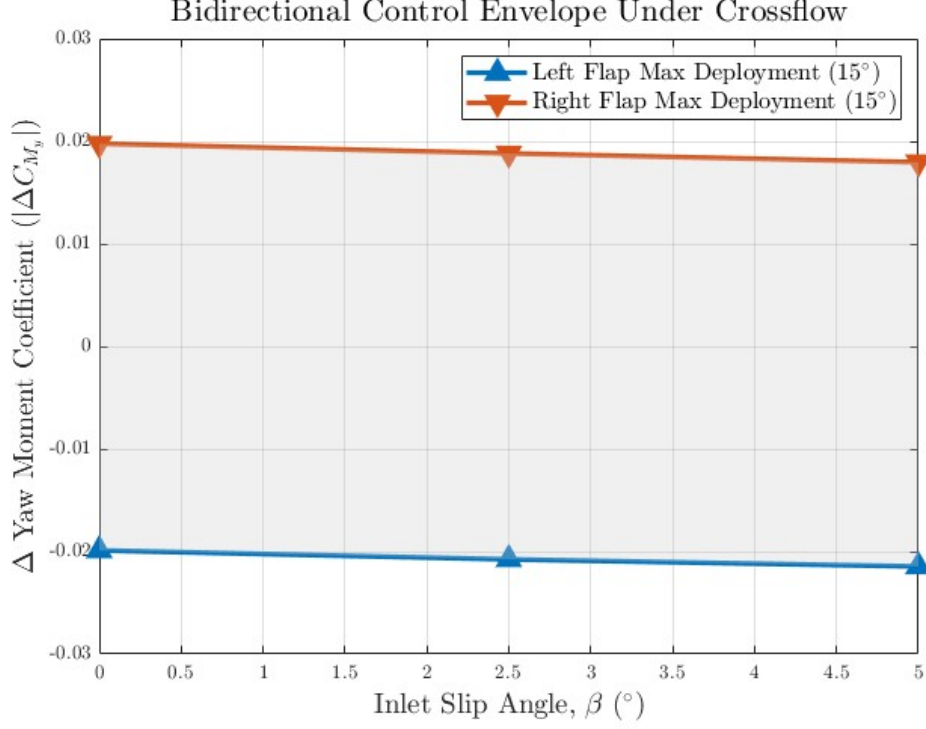


Figure 10: Bidirectional control authority envelope across inlet slip angles up to 5° . The shaded region represents the total spectrum of aerodynamic yaw available from the isolated aerodynamic assembly at the simulated conditions. (The open right-flap marker at $\beta = 0^\circ$ is inferred by mirror symmetry; all remaining markers represent directly simulated cases.)

The shaded region between the two deployment extremes defines the total spectrum of aerodynamic yaw bounded by the two simulated deployment directions.

- **Windward Actuation (Left Flap):** Deploying into the crossflow increases the local stagnation pressure, amplifying yaw authority to $\Delta C_{M_y} = -0.0215$ at $\beta = 5^\circ$.
- **Leeward Actuation (Right Flap):** Even under the imposed crossflow on the leeward side, the dihedral geometry maintains authority, $\Delta C_{M_y} = +0.0180$ at $\beta = 5^\circ$.

This persistent envelope shows the ADVC maintains control authority under crossflow.

5.3 Parametric Scaling and Physical Moment Generation

The yaw moment exported from Fluent is already evaluated about the proposed vehicle centre of gravity and must therefore not be multiplied by an additional vehicle-scale lever arm. From Equation (11), the incremental yaw moment may be written as

$$\Delta M_y = \Delta \left[\int_A (r_z dF_x - r_x dF_z) \right] = \Delta M_{y,F_x} + \Delta M_{y,F_z}. \quad (22)$$

The expected side-force contribution can be estimated using a physically plausible longitudinal moment arm,

$$\ell_x = x_{F_z} - x_{CG}, \quad (23)$$

where x_{F_z} denotes the effective streamwise centre of action of the incremental side force. Both x_{F_z} and x_{CG} are coordinates in the global reference system, whereas ℓ_x represents their relative longitudinal separation. With $x_{CG} = -1.5$ m and the wing extending approximately from $x = 0$ to 0.30 m, the plausible geometric range is

$$1.5 \text{ m} \lesssim \ell_x \lesssim 1.8 \text{ m}. \quad (24)$$

For the following conservative estimate, the lower-bound value $\ell_x = 1.5$ m is adopted, corresponding to the assumption that the effective incremental side force acts at $x_{F_z} = 0$. The resulting quantity is therefore an estimated side-force-induced yaw contribution rather than an exact decomposition of the Fluent-integrated moment. It is given by

$$\Delta M_{y,F_z}^{\text{est}} = -\ell_x \Delta F_z, \quad \ell_x = 1.5 \text{ m}, \quad (25)$$

and its nondimensional form is

$$\Delta C_{M_y,F_z}^{\text{est}} = \frac{\Delta M_{y,F_z}^{\text{est}}}{q_\infty S_{\text{ref}} b_{\text{ref}}} = -\frac{\ell_x}{b_{\text{ref}}} \Delta C_{\text{side}}. \quad (26)$$

At zero sideslip, an apparent effective lever arm can be defined as the diagnostic ratio

$$\ell_{\text{eff}} \equiv -\frac{\Delta M_y}{\Delta F_z}. \quad (27)$$

Because ℓ_{eff} is calculated from the total yaw-moment increment, which includes both the longitudinal-force and side-force contributions in Equation (22), it should be interpreted as a diagnostic apparent lever arm rather than a direct measurement of x_{F_z} . The resulting values are 1.666 m, 1.669 m, and 1.671 m for the 5°, 10°, and 15° deployments, respectively. These values are nearly constant and lie within the plausible geometric interval of 1.5–1.8 m.

At 15°, for example, the measured $\Delta F_z = 8.081$ N implies an estimated side-force-induced yaw-moment magnitude between 12.12 and 14.55 N m, while the directly computed total yaw-moment increment has a magnitude of $|\Delta M_y| = 13.507$ N m. The observed yaw increment is therefore consistent with the measured side force acting through a plausible longitudinal lever arm. This agreement supports, but does not uniquely prove, side-force-dominated yaw generation.

For a comparable configuration, the approximate side-force contribution scales as

$$|\Delta M_{y,F_z}| \approx \frac{1}{2} \rho U_\infty^2 S_{\text{ref}} \ell_x |\Delta C_{\text{side}}(\delta, \beta; \alpha)|. \quad (28)$$

For a fixed incremental side-force coefficient, the available authority therefore scales quadratically with velocity and linearly with reference area and longitudinal moment arm. The coefficient itself depends on deployment angle, sideslip, airfoil geometry, and operating incidence. These parameters cannot be increased independently because greater aerodynamic loading or deployment may introduce separation, pitch and roll coupling, hinge torque, and structural penalties. The present geometry should consequently be interpreted as an unoptimized proof-of-concept baseline for a future multi-objective design study.

5.4 Cross-Axis Isolation and the Decoupled Yaw Actuator

A desirable characteristic of an aerodynamic control device is to provide predictable vectoring authority. When a stability intervention necessitates an induced yaw moment (C_{M_y}), any simultaneous induction of longitudinal drag, pitch, or roll acts as a cross-axis disturbance. Therefore, the

Active Dihedral Vectoring Control (ADVC) is evaluated on its capacity to act as a decoupled yaw actuator, generating lateral control authority without contaminating the longitudinal, vertical, and rotational axes.

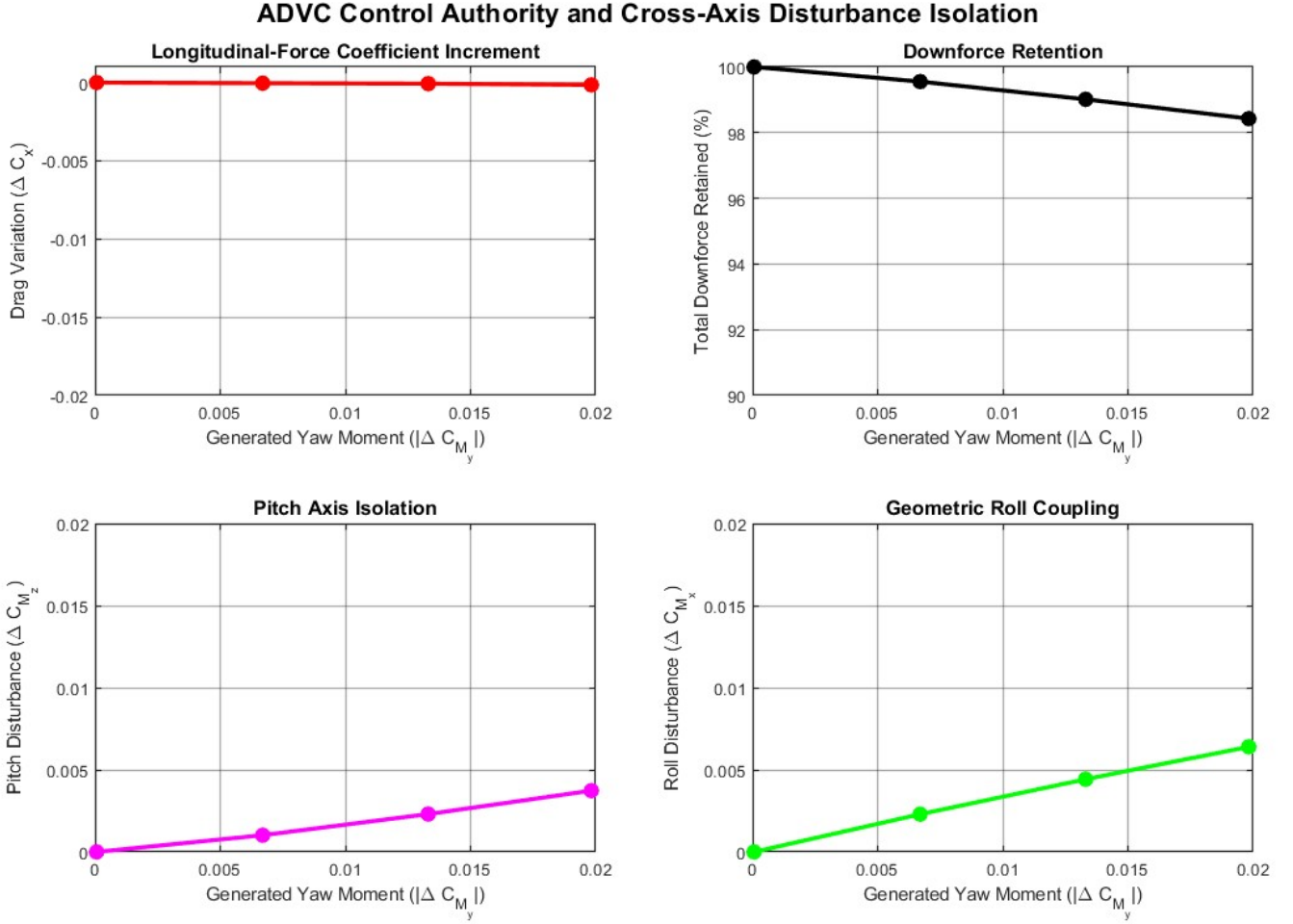


Figure 11: ADVC control authority and cross-axis response at zero sideslip. The horizontal axis presents the magnitude of the generated yaw-moment increment, $|\Delta C_{M_y}|$, because left-side deployment produces a negative yaw increment under the adopted right-hand-rule convention. The longitudinal, pitch, and roll quantities retain their signed definitions. Increasing $|\Delta C_{M_y}|$ corresponds to increasing left-flap deployment from $\delta_L = 0^\circ$ to 15° .

Longitudinal and Vertical Axis Isolation

As illustrated in Figure 11, the progression of the generated yaw moment corresponds directly to increasing ADVC deployment. The drag variation remains negligible over the investigated deployment range. This small longitudinal-force coupling suggests that, in a hypothetical full-vehicle implementation, the mechanism could potentially provide restorative yaw inputs with limited concurrent loss of straight-line aerodynamic performance; however, vehicle-level behavior was not evaluated in the present study.

Regarding vertical load, the observed penalty is primarily a geometric consequence of the dihedral deployment rather than massive parasitic flow separation. As the outboard flap actuates to 15° , its local normal force vector is tilted. A portion of its previously vertical aerodynamic load is explicitly redirected into the lateral plane, reducing its downforce contribution as a trigonometric function of

the deployment angle. Despite this geometric shift, the ADVC retains 98.4% of the isolated wing assembly's zero-sideslip baseline downforce.

Pitch Isolation and Geometric Roll Coupling

Utilizing the uniform reference length for the roll moment provides a clearer perspective on the relative magnitude of the cross-axis disturbances. The numerical results indicate that the ADVC exhibits a minor nose-down shift ($\Delta C_{M_z} \approx 0.004$ at peak deployment), suggesting that pitch coupling remains small relative to the primary yaw-moment increment.

While longitudinal and vertical variations are minimized, the lateral nature of the yaw generation inherently induces a roll moment (ΔC_{M_x}). However, as illustrated in the geometric roll coupling data, this disturbance is substantially smaller than the primary yaw authority, peaking at approximately $\Delta C_{M_x} \approx 0.006$ for a generated yaw moment of 0.020.

The smaller roll increment is consistent with the shorter vertical side-force lever arm relative to the longitudinal yaw lever arm. At 15° , using the approximate vertical offset of 0.6 m gives a side-force roll estimate of $0.6(8.081) = 4.85$ N m, compared with the computed total roll increment of 4.35 N m. The difference is consistent with partial opposition from vertical-load redistribution (since the deployed flap would produce less downforce) and other surface-force contributions, although these components were not independently decomposed.

6 Flow Visualization and Qualitative Aerodynamic Assessment

To provide qualitative context for the integrated loads of the Active Dihedral Vector Control (ADVC) mechanism, a comprehensive 3D flow visualization study was conducted. The primary objective was to ensure that the deployment of the 15° dihedral flap generates the required lateral yaw authority without inducing macroscopic boundary layer separation or catastrophic loss of baseline downforce.

6.1 Global Flow Attachment and Surface Pressure

A key consideration for the studied ADVC architecture is its ability to maintain organized flow within the freestream. Figure 12 presents the global velocity pathlines over the wing assembly with the outboard left flap deployed to 15° . The visualized streamlines suggest general conformity to the aerodynamic surfaces, without showing obvious gross wake disruption.

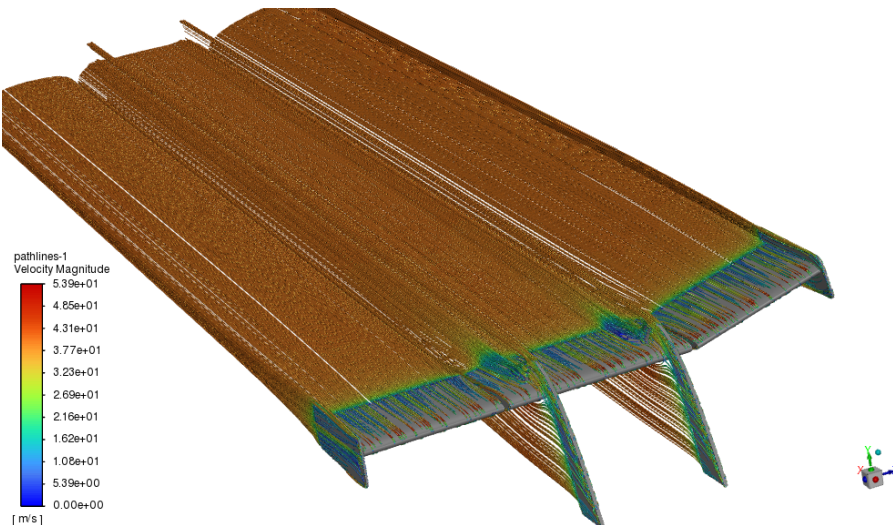


Figure 12: Global velocity pathlines over the ADVC assembly (15° left deployment).

As is shown in the comparative underside static pressure contours (Figure 13), The corresponding pressure contours remain broadly similar between the baseline and deployed configurations, providing qualitative context for the observed downforce retention.

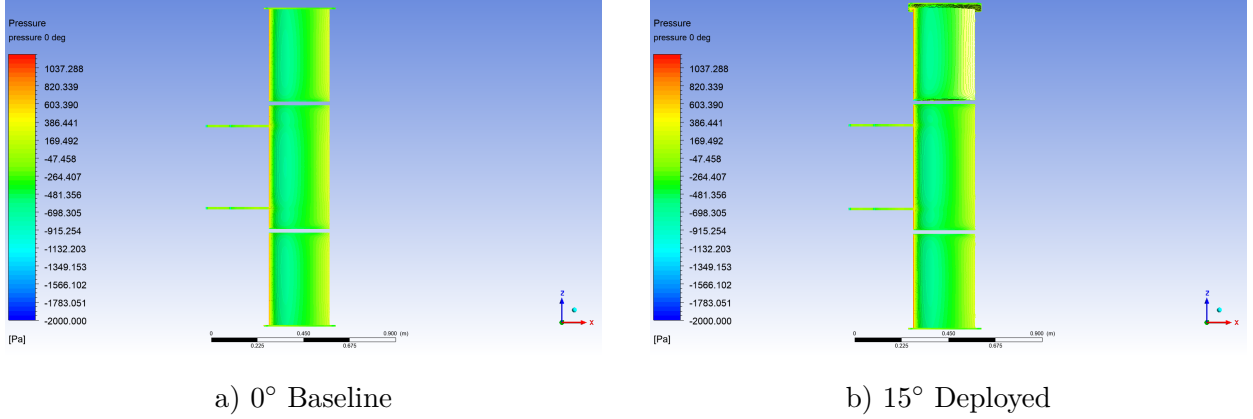


Figure 13: Comparison of underside static pressure contours, showing the preservation of primary downforce structures.

6.2 Wake Topology and Parasitic Vortex Generation

While the global streamlines indicate broad surface attachment, the wake topology was further investigated using the Q-Criterion method introduced by Hunt, Wray, and Moin [8] to identify regions where rotational kinetic energy dominates fluid strain.

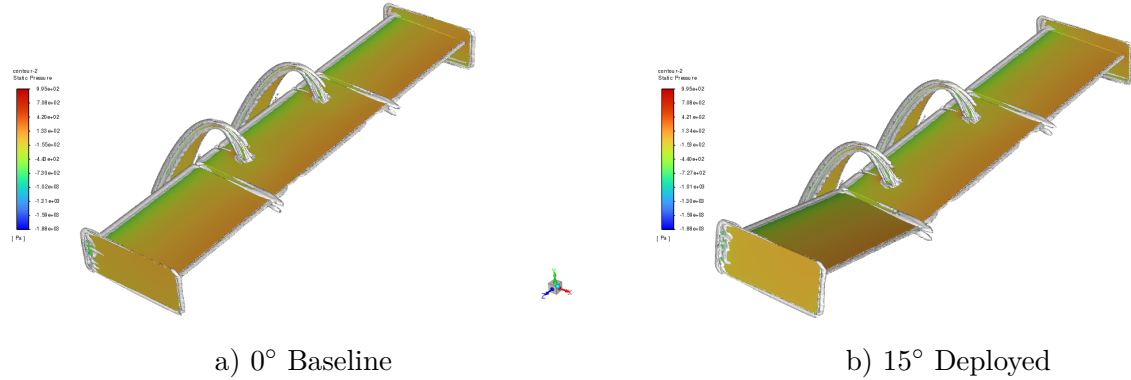


Figure 14: Iso-surfaces of the Q-criterion at $Q_{th} = 5.0 \times 10^5 \text{ s}^{-2}$, overlaid on wing-surface static-pressure contours for (a) the baseline and (b) the 15° deployed configuration. Identical threshold, pressure-contour range, camera orientation, and visualization settings were applied to both cases.

At the selected Q-criterion threshold, the displayed large-scale wake structures remain broadly similar between the baseline and deployed configurations. This qualitative similarity is consistent with the small resolved longitudinal-force coefficient increment (ΔC_x) and supports the interpretation that the lateral force (F_z) is generated primarily through the localized reorientation of the aerodynamic loading (i.e., dihedral deployment) rather than gross flow separation or disruption. Figure 14 is therefore used only for a qualitative comparison of the displayed wake topology and is not treated as an independent measure of vortex strength, drag, or boundary-layer attachment.

6.3 Sectional Static Pressure Distribution

To provide a local load comparison, localized chordwise pressure data was extracted at the midpoint of the deployed flap ($Z = 0.525m$).

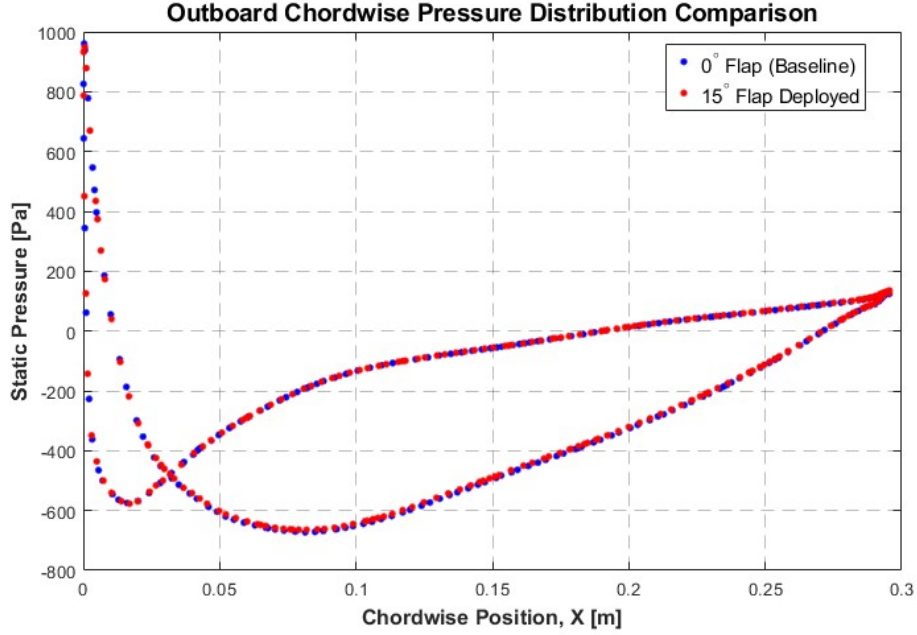


Figure 15: Chordwise pressure distribution comparison at the outboard section.

At the sampled outboard station, the baseline and deployed pressure distributions remain closely similar over most of the chord. This local comparison is consistent with limited alteration of the sectional aerodynamic loading. However, a single sectional pressure distribution does not independently quantify global drag or establish boundary-layer attachment.

6.4 Numerical Limitations and Evidence Scope

The present results should be interpreted as a preliminary comparative RANS investigation rather than a grid-independent aerodynamic validation. A formal mesh-refinement study was not performed, and the near-wall discretization was restricted to three prismatic layers by the available computational memory. The reported maximum $y^+ \approx 78$ characterizes only the upper end of the wall-resolution distribution and does not demonstrate that every surface location lies within a single near-wall regime. In addition, the simulations employed a steady RANS solver; therefore, residual plateaus and bounded oscillations cannot be uniquely attributed to physical wake unsteadiness without a transient calculation. The pathlines, pressure contours, and Q-criterion iso-surfaces are consequently used as qualitative evidence supporting the integrated load trends rather than as independent validation of attachment, vortex strength, or drag. Finally, the isolated-wing domain excludes the vehicle body, wheels, moving ground, suspension response, and wing-body interference. The most defensible outcomes of the study are therefore the direction, approximate magnitude, and repeatability of the incremental aerodynamic loads within the simulated configuration. Sub-percent longitudinal-force changes and full-vehicle handling benefits remain outside the established numerical accuracy and scope.

7 Conclusion

This exploratory study computationally investigated the aerodynamic characteristics of an Active Dihedral Vector Control (ADVC) mechanism for high-performance motorsport applications. The primary objective was to evaluate whether independently actuated dihedral outer surfaces could generate measurable yaw authority primarily through lateral side force, without severely penalizing longitudinal drag or global downforce.

Numerical simulations utilizing the steady Reynolds-averaged Navier–Stokes (RANS) CFD suggest that the theoretical ADVC mechanism can approximate this decoupled force response. By deploying the outboard flap to a 15° dihedral angle at zero sideslip, the system generates a directed lateral force (F_z) that induces the targeted yaw moment (C_{M_y}).

Furthermore, global flow visualizations and static pressure distributions indicate that this lateral vectoring is achieved with no obvious qualitative disruption to the general flow. The high-pressure stagnation zones and low-pressure suction peaks on the mainplane remain relatively undisturbed, resulting in a marginal downforce deviation of approximately 1.6%. These conclusions remain limited to the isolated-wing, steady-RANS configuration investigated.

Ultimately, this investigation indicates that the ADVC serves as an aerodynamically plausible mechanism warranting future full-vehicle and coupled-dynamics investigations. By preserving the general baseline wake topology, the proposed system demonstrates potential for decoupled yaw authority, providing a step for further optimization design studies.

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A Exported Test Cases

The following tables detail the raw force monitor outputs.

Table 2: Dimensional outputs from all 14 cases. Angles are in degrees; forces in N; moments in N m.

Case	δ_L	δ_R	β	F_x	F_{down}	F_z	M_y	M_z	M_x
DP 0	0.01	0.01	0.000	19.251	109.420	-0.025	0.041	-191.890	-0.029
DP 1	5.00	0.01	0.000	19.235	108.932	2.730	-4.549	-191.198	1.532
DP 2	10.00	0.01	0.000	19.222	108.342	5.418	-9.041	-190.325	2.977
DP 3	0.01	0.01	2.493	19.257	109.268	2.302	-2.877	-191.639	-
DP 4	10.00	0.01	2.493	19.201	108.514	7.947	-12.298	-190.574	-
DP 5	0.01	10.00	2.493	19.220	107.824	-2.845	5.741	-189.497	-
DP 6	0.01	0.01	5.005	19.250	109.073	4.644	-5.808	-191.314	-
DP 7	10.00	0.01	5.005	19.160	108.610	10.467	-15.510	-190.704	-
DP 8	0.01	10.00	5.005	19.244	107.269	-0.283	2.447	-188.610	-
DP 9	15.00	0.01	0.000	19.187	107.698	8.056	-13.466	-189.356	4.320
DP 10	15.00	0.01	2.493	19.171	108.129	10.760	-16.980	-190.007	-
DP 11	0.01	15.00	2.493	19.196	106.974	-5.334	9.903	-188.193	-
DP 12	15.00	0.01	5.005	19.146	108.337	13.377	-20.378	-190.310	9.297
DP 13	0.01	15.00	5.005	19.228	106.176	-2.628	6.389	-186.926	-